



Deciphering the Digital Noise: Airline Sentiment Analysis

Course: Concept & technology of AI

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1. Introduction

1.1 Problem Statement

In the age of social media, a single tweet can define a brand's reputation. For airlines, understanding customer sentiment isn't just about PR—it's about operational efficiency. This project aims to build an AI system that can automatically categorize tweets as **Positive, Neutral, or Negative**, allowing companies to respond to crises in real-time.

1.2 The Dataset

The dataset consists of thousands of tweets directed at major airlines. Each record contains the text of the tweet, the airline being mentioned, and the pre-labeled sentiment. It is a classic example of "unstructured data" that requires significant cleaning before a machine can understand it.

1.3 Alignment with UNSDG

This project aligns with **UNSDG Goal 9**. By automating the analysis of consumer feedback, we foster innovation in the service industry and build resilient digital infrastructures that allow businesses to grow through better customer relations.

2. Methodology

2.1 Pre-processing: From Text to Numbers

Raw tweets are noisy, filled with @mentions, hashtags, and emojis. My first step was a "deep clean"—stripping out special characters and converting the text into a format the models could digest. I utilized **TF-IDF Vectorization**, which essentially weighs words by their importance, ensuring that common words like "the" don't drown out meaningful words like "delayed" or "excellent."

2.2 Exploratory Data Analysis (EDA)

My initial exploration revealed a harsh reality: the vast majority of tweets directed at airlines are **Negative**. This "Class Imbalance" is a major challenge in AI, as models can easily become biased toward the majority class.

2.3 Feature Engineering

Beyond the text itself, I looked at metadata. Does the specific airline matter? Does the length of the tweet indicate the intensity of the sentiment? By combining these features with the vectorized text, I created a more holistic "view" for the models.

2.4 Model Building

I tested two distinct paths:

1. **The Neural Network (MLP):** A complex model designed to find hidden patterns in word associations.

2. **Classical Machine Learning:** I trained **Logistic Regression** and **Random Forest** models. Interestingly, for text data, these "simpler" models often outperform complex ones because they are less prone to overfitting on slang or typos.

2.5 Optimization

To ensure the **Logistic Regression** model wasn't just lucky, I used **GridSearchCV**. This allowed me to fine-tune the "regularization" parameters—essentially teaching the model to be strict enough to find patterns but flexible enough to work on new, unseen tweets.

3. Results and Conclusion

3.1 Key Findings

The competition between models was tight. While the Neural Network was powerful, the **Logistic Regression** model proved to be the most consistent. It maintained a high **F1-Score**, which is a crucial metric when dealing with imbalanced data because it balances both precision and recall.

3.2 Final Model Selection

I chose **Logistic Regression** as the final model. It achieved an accuracy of approximately **72.4%**. While that might sound lower than the regression task, in the world of human language and sarcasm, this is a very strong result.

3.3 Challenges

Sarcasm and Ambiguity were the biggest hurdles. A tweet saying "Great, another 4-hour delay!" contains the word "Great," which might confuse a simple model into thinking it's positive. Navigating these linguistic nuances remains the "final frontier" for this project.

4. Discussion

4.1 Performance Interpretation

The model is particularly good at identifying negative sentiment (where the data is plentiful) but struggles slightly more with neutral tweets, which are often the most ambiguous for AI to interpret.

4.2 Impact of Feature Selection

Removing "stop words" and focusing on high-impact terms was a game-changer. It reduced the noise in the dataset and allowed the model to focus on the emotional "weight" of the tweets.

4.3 Limitations and Future Research

Currently, the model treats every word as an individual unit. For future work, I recommend using **Word Embeddings (like Word2Vec)** or **Transformers (like BERT)**. These would allow the AI to understand the *context* and *sequence* of words, potentially solving the sarcasm problem.

FIGURE

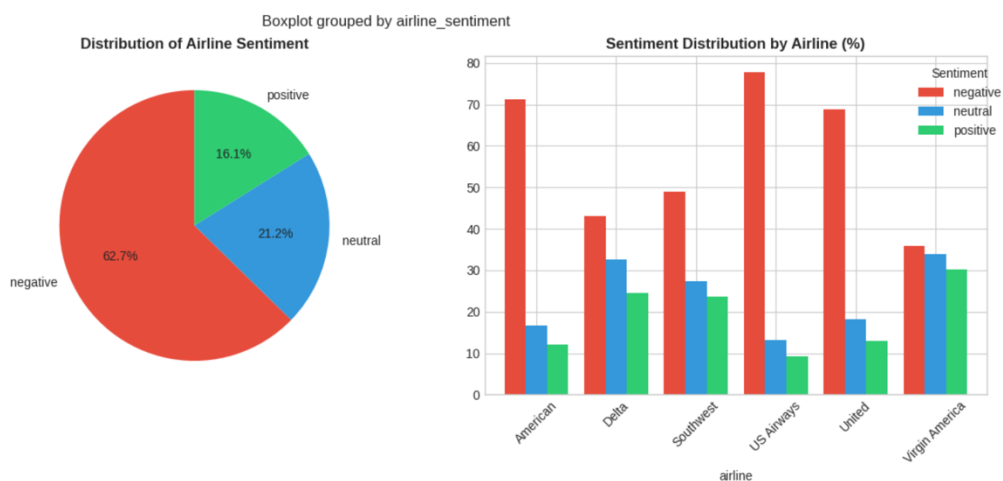


FIG. Distribution of Airline Sentiment & Sentiment Distribution by Airline

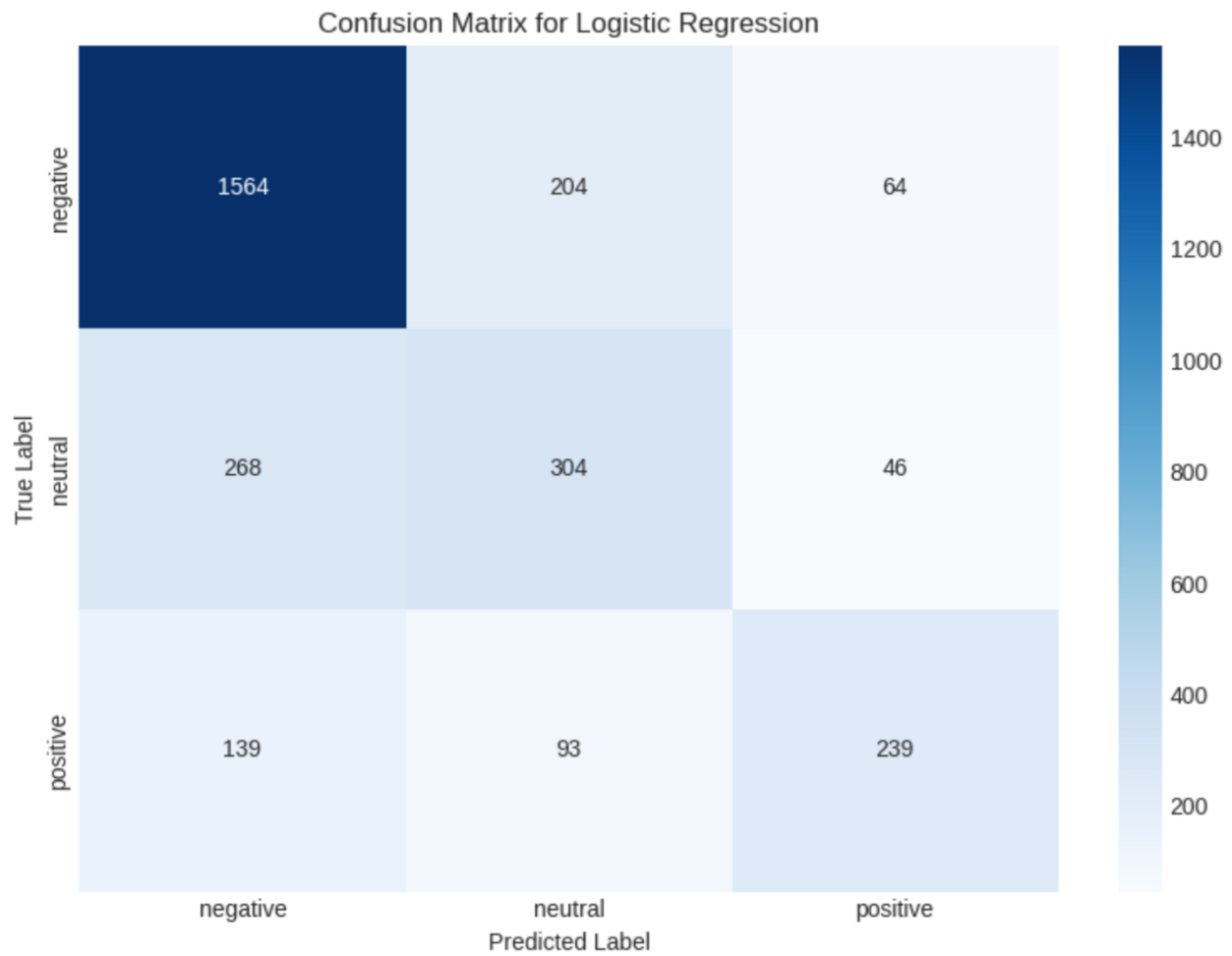


Fig. Confusion Matrix for Regression

5. References

- Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*.
- Scikit-learn: *Machine Learning in Python*, Pedregosa et al., JMLR 12, 2011.
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